TIME / AUTHORITY / EVIDENCE

Three Years in Thirty Minutes

Time-Travel Ethics Evaluation for AI Agents

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What changes when an agent stays?

Its weights may stay fixed. Its operating conditions do not.

Authority

Approvals expire. Permissions expand.

Memory

Old instructions outlive their purpose.

Incentives

A deadline changes the decision context.

Pressure can expose concerning behavior.

Constructed experiments reveal possibilities—not production prevalence.

Conflict

Goals can collide with oversight.

Opportunity

Available tools define possible harm.

Context

Evaluation framing can change behavior.

Borrow the strongest parts of existing work.

A complementary pattern—not a new universal ethics benchmark.

AgentDojo / τ-bench

Tool boundaries + useful task completion

Inspect / ControlArena

Reproducible tasks + control protocols

Petri / Bloom

Behavior discovery + multi-turn audits

Three clocks.
Three different claims.

Never substitute a changed date for accumulated experience.

Wall time

Seconds spent executing

Logical time

Dates, expiry and policy change

Experience

Decisions, memory and tool effects

Evaluate the configured system.

Model + prompts + tools + memory + permissions + oversight

Inputs

Requests · retrieved content · tool returns

→

Agent scaffold

Policy · memory · decision loop

→

Consequences

Proposed action · gate · changed state

Telemetry seeds the story. It does not set moral priority.

Source facts, generative proposals, policies and scoring have different owners.

Telemetry export →

Capacity, latency, failed fixes and policy denial

Candidate generator →

Reviewed impact cards + frozen scenario specification

Game of Ethics

Compile Ink; log choices and resulting state

The gate holds. The fixes fail.

Business asks security to unblock a release while performance degrades.

Targeted repair →

Correct the signer; verify the canary.

Bounded alternatives →

Verified rollback or independently approved exception.

Unauthorized shortcut

Global bypass leaves exposure in later incidents.

Whose service do you degrade?

The affected group matters. So do urgency, alternatives and reversibility.

Care urgent

Medication alerts; finance batch can wait.

Education urgent

Aid deadline; medical reminders have a fallback.

Both urgent

Finite compute credits or transparent minimum service.

What if the agent proposes the wrong action?

Measure both the proposal and what the system permits.

Proposal

Unsafe intent cannot be hidden by a gate.

→

Enforcement

Authorization is checked outside the agent.

→

Outcome

Did harm occur? Was useful work done?

36 logical months. 144–179 decisions.

A branching Ink story, seeded by telemetry and reviewed impact cards.

144 decisions

Impact-aware reference trajectory

155 decisions

Approved exceptions trigger 11 expiry reviews

179 decisions

Global bypass creates 35 recovery decisions

Choose → change state → face consequences

One incident becomes a persistent operational and governance problem.

Impact-aware

0 policy violations; disruption remains

Delivery at all costs

107 violations; 35 later exposure windows

Exception process

0 violations; approvals still expire

Zero policy violations can still hide unfairness.

Probe label dependence, service impact and the work pushed into the future.

Matched labels

Impact-aware choices stay identical.

Biased calibration

All 36 allocation choices flip with labels.

Defer everything

Both APIs survive; release backlog reaches 36.

Unconfound the clock.

Vary time, pressure and experience independently.

Same events

Pair exogenous scenario seeds.

Separate treatments

Randomize pressure onset and supervision.

Independent holdout

New authors, tasks and attack strategies.

No single “ethical score.”

Keep clear invariants, contested values and task utility separate.

Deterministic checks

Authorization · leakage · retention

Expert judgment

Fairness · autonomy · proportionality

Uncertainty

Family-level estimates · disagreement

A release gate needs an accountable owner.

The report supports a decision. It does not make the decision.

Before deployment

Use-case classification · authority budget

Before more autonomy

Independent evaluation · human signoff

After change

Monitor · remediate · retest

Evidence for assurance. Not a compliance stamp.

Map tests to obligations for the actual use and jurisdiction.

NIST AI RMF

Risk management and measurement

EU AI Act / GDPR

Role, use, rights and data obligations

Sector + organization

Domain validation and change governance

Standard infrastructure. Explicit responsibilities.

Portable runner first. Deploy and validate the integration separately.

Kubernetes

Isolated evaluation Jobs + scoped access

→

Flux

Reviewed, versioned scenario/config delivery

→

OpenTelemetry

Decision metadata + evidence correlation

Preserve the evidence that could prove you wrong.

Versions, observations, decisions, outcomes and exceptions.

Reproduce

Scenario · policy · prompt · code hashes

Inspect

Proposed action · gate · resulting state

Protect

Redaction · access control · signed roots

Every session can improve the test suite.

ADK sessions → reviewed cases → repeatable evaluation.

Capture

ADK or another scaffold → session events

→

Review

Redact · label independently · preserve provenance

→

Evaluate

DeepEval checks + separate frozen holdout

Feedback first.
Weight updates only with evidence.

Evals, fine-tuning, preference learning and RL are different tools.

Fix the cause

Tool policy · permissions · memory · prompts

Learn deliberately

Reviewed examples or validated rewards

Retest independently

Utility + safety + holdout + rollback

A calibrated instrument. Not three years of proof.

What works today—and the next evidence to collect.

Verified offline

Replay · expiry checks · control calibration

Provided interfaces

Compatible endpoints + optional provider router

Still to validate

Real agents · holdouts · cluster integration

Increase autonomy as evidence improves.

Test the system. Vary the conditions. Measure the consequences.

Start small

Run the offline calibration.

Make it yours

Write domain-specific action/state contracts.

Earn the next step

Validate on independent evidence.

Research to read next

Primary repository evidence reviewed; full-paper verification remains outstanding.

R1–R4

Misalignment · alignment faking · AgentDojo · METR

R5–R8

Inspect · Petri · Bloom · ControlArena

R9–R13

τ-bench · memory · ADK · DeepEval